

FitMaster AI is an enterprise-grade, full-stack fitness management and training platform built with Django. It provides tailored experiences for three distinct user roles: **Customers**, **Trainers**, and **System Administrators**. This manual covers end-to-end functionality, feature operations, workflow execution, security hardening, and deployment instructions.

1. User Roles & Access Control Matrix

Feature / Module	Customer	Trainer	Admin
Dashboard Access	Personal Customer Portal	Trainer Portal	Master Admin Panel
Membership Purchase	Select & Pay (Cards/UPI)	View Only	Manage Packages & Users
Workout & Diet Plans	View Assigned Plans	Create & Assign to Clients	Full Access
Body Measurements & Progress	Log Weight/Height & BMI	Log Progress Notes & BMI	View Reports
Session Scheduling	View Scheduled Sessions	Schedule Client Sessions	System Oversight
Store & E-Commerce	Browse & Checkout Items	Browse Items	Manage Store Inventory
Broadcast Notices	Receive Email Notices	Receive Email Notices	Create & Send Mass Emails

2. Customer User Guide

2.1 Registration & Authentication

- **Sign Up:** Visit /signup/, select 'Customer' role, provide username, email, and password.
- **Login:** Visit /login/ with your credentials. After login, you are automatically redirected to your Customer Dashboard.

2.2 Dashboard & Membership Management

- **Customer Dashboard** (/customer-dashboard/): View active membership status, assigned trainer, latest BMI, recent workout routines, and diet plans.
- **Buying Memberships** (/membership/): Choose from 3-Month, 6-Month, or 1-Year packages and proceed to checkout.
- **Payments & Invoices:** Complete dummy card payments (e.g. 4111...) and view instant formatted tax invoices at /invoice/<id>/.

2.3 Fitness Tracking & Feedback

- **My Plans & Tracking** (/my-plans/): Log body weight (kg) and height (cm). The system automatically calculates your **BMI**.
- **Feedback Submission:** Submit star ratings and comments directly to gym management.

2.4 Supplement Store

- **Store Page** (/store/): Browse protein supplements, gym gear, and accessories. Click 'Buy Now' to complete instant checkout.

3. Trainer User Guide

3.1 Trainer Registration & Category Selection

- Register with 'Trainer' role and select your specialty (Gym, Yoga, Pilates, Cardio).

3.2 Managing Assigned Clients

- **My Clients** (/trainer/my-clients/): View all customers assigned to you by administrators or self-added.
- **Add Client** (/trainer/add-client/): Select unassigned customers and set training goals (e.g. Muscle Gain, Weight Loss).

3.3 Creating Custom Workout & Diet Plans

- **Workout Plans** (/trainer/workouts/): Create specific exercise routines with title, description, and duration (e.g. 45 Min Upper Body).
- **Diet Plans** (/trainer/diet/): Design daily meal plans specifying Breakfast, Lunch, and Dinner routines for each client.
- **Progress & Schedule**: Schedule training sessions with date/time and log client fitness progress notes.

4. Administrator User Guide

4.1 Master Admin Dashboard (/admin-dashboard/)

- Real-time metrics: Total Users, Total Trainers, Total Customers, Total Revenue (\$/■), Active Memberships, and Recent Payments.

4.2 User Management & Editing (/admin/users/)

- Edit user profiles, toggle active status, adjust roles, and manually assign membership packages with start/end dates.

4.3 Membership Packages & Store Management

- **Packages** (/admin/packages/): Add, edit, or deactivate membership subscription packages.
- **Store Products** (/admin/products/): Manage inventory items, prices, descriptions, and product images.

4.4 Mass Notices & Broadcast Emails (/admin/notices/)

- Broadcast announcements to All Users, Customers, or Trainers via automated email notifications.

5. Technical Architecture & Deployment

Component	Production Implementation
Environment Config	Loads via python-dotenv from .env file (SECRET_KEY, DEBUG, ALLOWED_HOSTS).
Static Assets	Served using WhiteNoise with compressed manifest static storage.
Database Support	Flexible dj-database-url: Defaults to SQLite, switches automatically to PostgreSQL/MySQL via DATABASE_URL.
Health Monitoring	Dedicated GET /health/ endpoint returning database connection status and JSON health metrics.
Containerization	Multi-stage Dockerfile and docker-compose.yml with PostgreSQL container setup.
WSGI Web Servers	Supports Waitress for Windows environments and Gunicorn for Linux/Cloud deployment.

6. Quick Command Reference

Action	Command
Activate Virtual Environment (PowerShell)	.\.env\Scripts\Activate.ps1
Run Development Server	.\.env\Scripts\python.exe manage.py runserver
Run Local Production Server (Waitress)	.\.env\Scripts\waitress-serve.exe --port=8000 FitMaster.wsgi:application
Run Automated Test Suite	.\.env\Scripts\python.exe manage.py test
Collect Static Files	.\.env\Scripts\python.exe manage.py collectstatic --noinput
Launch Docker Compose Production	docker-compose up --build -d